BloodLine – Predictive Blood Shortage and Smart Donor Matching Platform

Project Proposal for UCS503P
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1 Higher-order goal *...secondary framing*

This project aims to improve blood availability by reducing the time and coordination effort required to identify and respond to potential blood shortages. While the topic is broader than any single blood bank, the immediate engineering objective is to translate *proactive blood availability management* into a measurable, deployable software solution.

2 Time-to-value *...primary heuristic*

- We will deliver meaningful increments quickly by prototyping the core user workflow of inventory tracking, demand forecasting, shortage-risk detection, and donor matching within a short iteration window.
- We will prioritize fast feedback over perfection by using weekly iterations (and targeting CI-backed automation early) to reduce release friction.
- Success is defined by what can be delivered and measured in the first iteration, then improved based on testing and user feedback.

3 Problem Statement

Blood banks and hospitals can face uncertainty and delays in maintaining adequate blood availability. Common pain points include:

- **Reactive shortage management:** current inventory can be monitored, but short-term future shortage risk requires forecasting and early warning.
- **Fragmented coordination:** inventory, nearby blood-bank availability, hospital requirements, and potential donors are not unified into one workflow.
- **Inefficient donor outreach:** finding suitable nearby donors based on blood group, donation history, proximity, availability, and consent can be time-consuming.

Why this matters now (contextual relevance): A proactive, data-driven coordination layer can help blood-bank staff prepare for potential shortages earlier and reduce avoidable delays in donor outreach or cross-bank coordination.

4 Proposed Solution *...Software Proposition*

4.1 Overview

Build a web-based "BloodLine" system with the following components:

- **Inventory dashboard:** authorized blood-bank staff record and monitor blood stock by blood group/component.
- **Demand forecasting:** estimate near-term demand from historical data, initially using a simulated dataset where suitable granular real data is unavailable.
- **Shortage-risk detection:** compare projected demand with available/expected inventory and flag potential shortages.
- **Smart donor matching:** identify potential nearby donors using configured blood-group, cooldown, proximity, availability, and consent rules.
- **Role-based workflow:** provide separate interfaces for donors, blood-bank managers, inventory staff, medical/screening staff, hospital coordinators, and system administrators.

4.2 Core workflow

1. Blood-bank staff update current inventory and donation records.
2. BloodLine forecasts near-term demand and flags potential shortage risk.
3. The manager reviews the risk, nearby availability, and potential donor matches.
4. Donor outreach is initiated; medical staff perform final screening and the resulting donation is recorded.

4.3 Operational constraints for an educational, multi-user setting

- Keep the solution usable on low-to-mid range devices via responsive web design.
- Use clearly labelled simulated historical demand data where granular real consumption data is unavailable for the prototype.
- BloodLine does not make final medical eligibility decisions; donor screening and actual redistribution remain with authorized blood-centre staff.

5 Solution Approach *...Engineering focus*

This is an engineering course project, so we focus on reliable multi-user workflow, role-based access, system correctness, and deployability rather than novel algorithm research.

5.1 Forecasting and donor-matching assistance *...non-research*

Demand forecasting will use a time-series approach such as Prophet or SARIMA. The model will estimate near-term demand from the available historical/simulated dataset. A transparent rule-based layer will compare projected demand with available inventory and generate a shortage risk signal. Potential donors will then be filtered using blood-group compatibility, donation cooldown, approximate location, availability, and notification consent. Final medical eligibility remains with qualified staff.

5.2 Web architecture *...web-based solution*

A typical 3-tier web architecture:

- **Frontend:** responsive UI for donors, blood-bank staff, hospitals, and administrators.
- **Backend API:** authentication, role-based authorization, inventory operations, forecasting, risk detection, matching, requests, and notifications.
- **Database:** store users, roles, blood banks, inventory, donations, demand history, predictions, requests, and alerts.

5.3 CI/CD and fast delivery enabler

CI/CD will be used to:

- Automatically build and run tests on every change.
- Produce repeatable deployment artifacts to staging.
- Enable safe and frequent releases of incremental features.

6 Evaluation Criterion *...measurable and attributable*

We define success using measurable metrics tied to the core workflow.

6.1 Primary evaluation metric

Forecasting error: compare predicted demand with actual values in the held-out test period using MAE and RMSE.

- **Attribution:** calculated from forecast outputs and corresponding observed demand values in the test dataset.

6.2 Secondary evaluation metrics

- **Shortage-risk performance:** measure whether known shortage-risk cases in the test data are correctly identified.
- **Donor matching correctness:** verify that configured blood-group, cooldown, proximity, availability, and consent filters are applied correctly.
- **Workflow response time:** measure response time for common operations such as inventory update, dashboard loading, and donor search.
- **Notification targeting:** verify that alerts are limited to donors satisfying the configured matching rules.

6.3 Pilot validation plan *...high-level*

- Use a simulated demand dataset and divide it chronologically into training and testing periods.
- Compare metrics before/after within the iteration window using forecast outputs and seeded donor/blood-bank scenarios.

7 Scalability *...with foresight*

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** The system should support growth in number of blood banks, donors, inventory records, and concurrent dashboard usage by:
   - decoupling components (frontend/backend),
   - enabling pagination and indexing for common queries,
   - using stateless API design.
2. **Operationally deployable (practically):** The system should run on common web hosting:
   - managed database,
   - platform-hosted deployment,
   - CI/CD pipeline for staging/production.

8 Engine availability heuristic

A mature set of "engines" (tooling/services) is available to implement this as a web-based project:

- React.js + Tailwind CSS for responsive web interfaces.
- FastAPI (Python) for REST APIs and application logic.
- PostgreSQL for structured relational data.
- Pandas with statsmodels/Prophet for demand forecasting.
- Google Maps API or equivalent for proximity and regional visualization.
- Authentication, notifications, testing framework, CI runner, and staging deployment target.

We will use these mature components to focus on workflow design, correctness, integration, and measurement rather than inventing new infrastructure.

9 Project scope and deliverables *...iteration-friendly*

9.1 Initial deliverable *...first iteration*

- Role-based login and dashboards.
- Blood-bank inventory and donor records.
- Simulated demand dataset and forecasting pipeline.
- Shortage-risk calculation and basic donor matching.
- Basic logging and timestamps.
- CI: build + backend unit tests + linting; CD to staging.

9.2 Subsequent deliverables

- Regional blood-availability heatmap and nearby surplus/shortage coordination.
- Targeted in-app/push notification workflow.
- Hospital blood-request and medical screening workflows.
- Improved forecasting evaluation and indexed queries for scale readiness.

10 Risks and mitigations

- **Data availability:** use clearly labelled simulated data for the MVP and keep the data layer ready for authorized real data.
- **Forecast uncertainty:** present predictions as risk signals rather than guarantees and retain human review.
- **Medical/privacy constraints:** final donor clearance remains with qualified staff; minimize personal data and restrict access by role.
- **Operational issues:** start with in-app/push notifications and use staging early for repeatable deployments.

11 Summary

BloodLine turns blood availability management from a reactive, manually-coordinated process into a proactive, data-driven workflow. By combining inventory tracking, short-term demand forecasting, shortage-risk detection, and rule-based smart donor matching within a single role-based platform, it closes the coordination gap between blood banks, hospitals, and donors that currently causes avoidable delays. The project is scoped for rapid, measurable delivery: an MVP built on a mature, well-supported stack, validated against clear forecasting and matching metrics, shipped through CI/CD, and designed from the outset to scale to more blood banks and higher concurrent usage. The result is an engineering-focused, deployable system that demonstrates sound workflow design and correctness while addressing a real problem in blood availability coordination.